

X3D First-Time Setup & Optimization Guide — Project Baseline

Status: Planning baseline v0.2 (decisions 1–4 resolved)

Distribution: NEW standalone repo. Pulls assets/concepts from two existing projects: the **iRacing X3D Tuning kit** (no6969el.github.io/iracing-x3d-tuning) and the **X3D Undervolt Tester**. Those projects link to this repo as the foundation/prequel; this repo links to them as the next steps.

Primary target: AMD Ryzen 9 9950X3D (dual-CCD, single V-Cache die)

Framework goal: Extensible to ALL X3D chips (single-CCD, dual-CCD single-cache, dual-CCD dual-cache)

Scope split: The iRacing kit covers per-application tuning (timer resolution, GPU interrupt affinity, Process Lasso CPU Sets). The Undervolt Tester covers CO validation workflows. THIS guide covers everything that happens BEFORE and AROUND those: BIOS foundation, memory training integrity, curve optimization on-ramp, Windows baseline, and validation tooling.

Audience floor: Total first-timer. One line on what BIOS is + how to access it (Del/F2 during boot on most PCs, vendor-specific keys in the translation table). No further

hand-holding beyond that — the guide's verification tooling is the real safety net.

0. Purpose Statement (for any model/agent reading this)

This document is the master plan for building a first-time setup guide that takes a user from "just built the PC" (or "never touched BIOS") to a verified, stable, repeatable X3D tune — ending exactly where the iRacing X3D Tuning kit picks up. The philosophy carried over from the existing kit:

1. **Verify, don't assume.** Every setting change gets a verification step (script or manual readout).
 2. **Repeatability over peak numbers.** A tune that trains identically every boot beats a fragile record run.
 3. **Auto-detect, then branch.** Chip detection drives which instructions apply. The detection script from the iRacing kit will be reused (DO NOT rewrite — asset will be supplied).
 4. **Tested or sourced.** Every claim is either (a) validated through our own troubleshooting history from the iRacing kit, or (b) cited from known-good community/vendor sources, or (c) explicitly marked `[VERIFY]` .
-

1. Chip Framework (Architecture Matrix)

The guide branches on three architectural dimensions.
Build all content against this matrix from day one, even though v1 ships 9950X3D-focused.

Class	Example Chips	CCDs	V-Cache Location	Key Tuning Implication
A: Single- CCD X3D	5800X3D, 7800X3D, 9800X3D	1	The only CCD	No scheduling problem. Tuning = memory, CO, thermals. Simplest path.
B: Dual- CCD, single cache	7900X3D, 7950X3D, 9900X3D, 9950X3D	2	CCD0 (Zen 5) [VERIFY per - SKU]	The hard case. Scheduling (cache CCD vs frequency CCD), core parking, driver stack, per- CCD CO

Class	Example Chips	CCDs	V-Cache Location	Key Tuning Implication
				limits. This is why the 9950X3D "needs tuning to shine."
C: Dual-CCD, dual cache	9950X3D2 (cache on both CCDs) [VERIFY specs/naming at intake]	2	Both	Scheduling pressure drops (both CCDs are cache CCDs), but cross-CCD latency still matters. CO behavior per CCD still asymmetric by silicon quality.

Detection: Reuse the existing auto-detection script from the iRacing X3D tuner (asset to be supplied). It should map any detected chip to a class (A/B/C) + per-SKU

profile. New chips get added as profile entries, not new code paths.

Per-SKU profile fields (draft schema):

- Model string / family / model / stepping
 - CCD count, cache CCD index, cores per CCD
 - Stock boost clocks per CCD [VERIFY per SKU]
 - Known-safe CO starting offsets (conservative)
 - SoC voltage guardrail
 - Known BIOS features relevant to the SKU (e.g., X3D Turbo Mode availability)
-

2. Phase 0 — BIOS Foundation

Everything here happens before Windows matters.
Order matters.

2.1 BIOS/AGESA baseline

- Update to a current, known-stable BIOS. Record AGESA version — memory training behavior and CO behavior change across AGESA releases.
[RESEARCH: pin a minimum recommended AGESA for AM5 Zen 5 X3D]
- Load optimized defaults after flashing (clears stale training data).
- **RESOLVED — Vendor handling:** Guide is written in generic setting names, backed by a **BIOS Translation**

Table covering the top 5 board vendors (ASUS, MSI, Gigabyte, ASRock, Biostar [CONFIRM 5th vendor choice]). User selects their board vendor once (or auto-detected via baseboard WMI query in the detect script) and all instructions render/adjust with that vendor's exact wording and menu paths. Every setting also gets an expandable "other names for this setting" dropdown as the fallback for unlisted boards/BIOS revisions. Menu paths noted per vendor where stable (e.g., ASUS: Ai Tweaker; MSI: OC; Gigabyte: Tweaker).

2.2 Memory: EXPO + guaranteed training integrity

This is a core pillar of the guide per the brain dump: **the memory tune must be trained, and must STAY trained, so the state is guaranteed identical at every boot.**

- Enable EXPO profile (or manual timings for advanced path).
- **Memory Context Restore (MCR): DISABLED.** With MCR enabled, the board skips full DRAM training on boot and restores cached training data — faster boots, but training state can drift or restore stale/marginal data, producing "it was stable yesterday" instability. Disabling MCR forces a full retrain every boot: slower POST, guaranteed consistent training. [VALIDATED-DIRECTION —

confirm against our own kit troubleshooting notes at asset intake]

- **Power Down Enable: DISABLED** (paired with MCR on most AM5 boards – both off for training consistency). [VERIFY pairing behavior per vendor]
- Record trained values after first successful boot (tRFC, actual UCLK/MCLK/FCLK) – these become the reference fingerprint the testing kit checks against.
- FCLK/UCLK/MCLK ratio guidance: 1:1 UCLK=MCLK for ≤ 6000 MT/s sweet spot; FCLK 2000–2133 typical. [RESEARCH: current consensus for Zen 5]
- **SoC voltage guardrail: $\leq 1.30V$ hard cap** (post-burnout AGESA-enforced, but verify the board respects it and don't manually exceed).
- Optional advanced section: manual subtimings, tRFC tightening, bank group swap – explicitly out of scope for "first-time setup," linked as a follow-on.

2.3 PBO & Curve Optimizer (negative offsets)

- Start philosophy: **every chip's silicon is different – start low, validate, step down.**
- **RESOLVED – Two-path structure:**
 - **Path 1 (default, everyone):** All-core conservative negative offset (–10 to –15 starting range),

validated with the standard testing tiers. Done.

Most users stop here with a meaningful gain.

- **Path 2 (opt-in "Tuning Path"):** A software suite of tuning options + recommendations with full logging — this is where the X3D Undervolt Tester project gets absorbed/extended. Per-core refinement, guided stepping, per-CCD awareness, and the dual-format output spec (Section 4.4) so every result is both human-readable and AI-consumable.
- Dual-CCD asymmetry (Class B/C): cache CCD and frequency CCD tolerate different offsets. Cache CCD typically runs lower voltage/frequency ceilings; frequency CCD cores often take deeper offsets. Guide must present per-CCD guidance, not one number.
- PBO limits (PPT/TDC/EDC): motherboard vs manual — first-time setup keeps these at defaults or motherboard-enforced; document what they are and how to read them.
- Boost clock override (+MHz): optional, later-stage.
- **Key teaching point:** an unstable CO offset often doesn't crash under load — it crashes at IDLE or light load (low-voltage boost states). Testing section must reflect this (Core Cyclor-style single-core boosting tests, idle soak).

2.4 Scheduling-relevant BIOS settings (Class B/C)

- CPPC / CPPC Preferred Cores: enabled (driver stack depends on it).
- Global C-States: enabled (parking depends on it)
[VERIFY current best practice – some guides disable, we should test]
- SMT: on by default; note X3D Turbo Mode (ASUS) / equivalent "game mode" toggles that disable SMT + CCD1 – document as a benchmark curiosity, not the recommended path. [DECISION NEEDED]
- iGPU: disable if discrete GPU present (removes a device from the interrupt/scheduling picture; we saw the iGPU cause confusion in community troubleshooting).

2.5 Boot integrity checklist (end of Phase 0)

- [] AGESA version recorded
 - [] EXPO active, MCR off, Power Down off
 - [] Trained memory fingerprint recorded
 - [] SoC voltage confirmed $\leq 1.30V$
 - [] CO offsets recorded per core/CCD
 - [] Three consecutive cold boots train to identical values [tooling opportunity: fingerprint-compare script]
-

3. Phase 1 — Windows Install & Baseline

Assumes fresh Windows 11 install (24H2+). [DECISION NEEDED: do we cover the install itself / debloat depth?]

3.1 The driver stack (Class B/C critical path)

The dual-CCD X3D scheduling story lives in software:

1. **AMD chipset drivers** — installs the 3D V-Cache Performance Optimizer driver + PPM provisioning. Without this, cache-CCD preference doesn't work. Version matters; record it.
 2. **Windows Game Mode + Xbox Game Bar** — game detection is what triggers cache-CCD preference and frequency-CCD parking. Counter-intuitive for optimizers who instinctively strip Game Bar out. **The guide must explicitly warn: do not debloat away Game Bar on Class B chips (unless going the manual-affinity route from the iRacing kit).**
 3. Power plan: Balanced is the AMD-intended plan for parking behavior (not Ultimate/High Performance). [VALIDATED-DIRECTION — align with what we proved in the iRacing kit testing]
- Verification step: confirm `amd3dvcache` service/driver present and running; confirm

preferred-core rankings visible (e.g., via provisioning package readout). [tooling opportunity]

3.2 Windows baseline settings

- HAGS (Hardware-Accelerated GPU Scheduling): document current recommendation + how to test both states. [RESEARCH]
- Core isolation / VBS / Memory Integrity: measurable overhead on some systems; document tradeoff honestly (security vs latency) rather than blanket "disable."
- Timer resolution: reference the existing iRacing kit — that's its territory. Link, don't duplicate.
- Background/debloat: minimal, reversible changes only. No registry-hack soup. Everything scripted = everything reversible.
- NVIDIA driver install (clean, no bloat) + note our documented MEMORY_MANAGEMENT 0x61941 BSOD history with iRacing/NVIDIA drivers — driver version hygiene section. [Pull from our forum PSA]

3.3 Handoff point

Phase 1 ends where the iRacing X3D Tuning kit begins: per-application CPU Sets, GPU interrupt affinity, Process Lasso config. The first-time-setup guide links out; no duplication.

4. Phase 2 – Testing & Validation Kit

The differentiator. Not "trust me," but "run this and see."

4.1 Stability validation (new-tool candidates + known tools)

What we're validating	Tool(s)	Notes
Memory training/stability	TestMem5 (anta777/1usmus profiles), Karhu, y-cruncher VT3	Community standard; define pass thresholds + durations
CO per-core stability	CoreCycler (single-core boost residency)	Catches idle/light-load CO instability that all-core tests miss
All-core load + thermals	y-cruncher, OCCT, Cinebench soak	Record clocks/temps as fingerprint
Idle stability	Idle soak protocol (overnight)	The classic CO failure mode
Latency sanity	AIDA64 / MLC / custom	Confirms 1:1 ratios delivering

What we're validating	Tool(s)	Notes
		expected latency
Scheduling correctness	Custom (build this)	Confirm game threads land on cache CCD, parking engages — extend concepts from the iRacing kit's verification tooling

4.2 New kit tools (build list — draft)

1. **Chip detect** — REUSE existing script (supplied asset).
Extend output to class A/B/C + profile lookup.
2. **BIOS state auditor** — reads what's actually live from Windows: memory speed/timings (where readable), SoC voltage, CO can't be read directly [RESEARCH: what's readable via WMI/HWINFO SDK/ryzen_smu], MCR state inference via boot-time training detection [RESEARCH: feasible?]
3. **Training fingerprint compare** — capture trained memory state, diff across boots, alarm on drift.

Directly serves the "guarantee the proper tune at any given time" requirement.

4. **Scheduling verifier** — is the V-Cache driver loaded, is Game Mode on, do preferred cores rank the cache CCD, does a test workload land correctly.
5. **One-shot health report** — runs all checks, outputs a single pass/fail report the user can share (also = our support/debug intake format, same as the iRacing kit's diagnostic approach).
6. **Backup/restore integration** — fold in concepts from Backup-Interactive-v2.ps1 (registry/profile capture) so a known-good state is always restorable.

Language/platform assumption: PowerShell, no-install, matching the existing kit's distribution style. [CONFIRM]

4.3 Testing protocol structure

- Tiered: Quick (30 min) / Standard (2–3 hr) / Paranoid (overnight)
 - Each BIOS change → minimum Quick tier before proceeding (change one thing at a time)
 - Final state → Standard minimum, Paranoid recommended
 - Every pass/fail logged with the fingerprint tools → builds the user's own tuning history
-

4.4 Dual-Format Output Spec (Human + AI) — CORE FEATURE

Every tool in the suite (testers, auditors, fingerprint compare, health report) outputs in two stacked layers:

Layer 1 — Human-readable report (top):

- Clean summary: what was tested, pass/fail, key numbers, plain-language interpretation, recommended next step.
- Optimized for quick scanning — the best possible output for a human to read.

Layer 2 — "Format for AI Troubleshooting" (beneath, clearly delimited):

- A fenced, copy-paste-ready block containing:
 1. **Embedded prompt header** — tells the receiving model its role and objective: what system this is, what was being attempted, what the tool observed, and what the AI needs to *solve / find out / locate / recommend*. Written so ANY model (cloud or local LM Studio) can pick it up cold with zero prior context.
 2. **Structured state dump** — chip profile (from detect script: model, class A/B/C, CCD topology), BIOS-relevant state (AGESA if known, memory config, fingerprint), CO offsets under test, test parameters, raw results/error codes, and recent-change history from the log.

3. **Explicit ask** — the specific question(s), e.g., "Core 7 failed CoreCycler at -25 after 40 min; all others stable at -30. Recommend next offset step for core 7 and whether the asymmetry indicates a preferred-core interaction."

- Design rule: the block must be self-contained. The user should never have to explain their setup — the tool already did.
- Logging: every run appends to a local history log (JSON or structured text [DECISION]) so the AI block can include "what's been tried already" — this is what makes iterative tuning sessions with a model actually work.

Why this matters: it converts every failure state from "go ask a forum and wait" into "paste this block into any model and continue tuning in the next five minutes." It also standardizes our own support intake.

5. Assets To Be Supplied (intake list)

- [] Auto-detection script from iRacing X3D tuner (reuse as-is / extend)
- [] **X3D Undervolt Tester** — full project (becomes the engine of the opt-in Tuning Path; assess what carries over vs. extends)
- [] Existing iRacing kit scripts + README (for cross-linking, tone/format consistency, and mining)

- validated findings)
- [] Troubleshooting notes/history from kit development (source of [VALIDATED] claims)
 - [] NVIDIA BSOD PSA content (driver hygiene section)
 - [] Backup-Interactive-v2.ps1 (backup/restore concepts)

6. Open Research Items (consolidated)

1. Minimum recommended AGESA for Zen 5 X3D + known-bad versions
2. MCR/Power Down pairing behavior per board vendor; whether MCR-off is universally the right call or "off until validated, then optionally on with fingerprint monitoring"
3. What tune state is programmatically readable from Windows (for the auditor tool)
4. Current Zen 5 consensus: FCLK sweet spot, UCLK=MCLK ceiling, 6000 vs 6400 guidance
5. C-states / parking interaction current best practice (test, don't copy forums)
6. HAGS recommendation with current NVIDIA drivers
7. 9950X3D2 (Class C) confirmed specs: cache CCD topology, clocks, whether the scheduling driver behaves differently [USER-SUPPLIED CLAIM – verify at intake]

8. Per-SKU cache-CCD index confirmation across all X3D chips (for the profile table)

7. Decisions

Resolved:

1. ~~Distribution~~ → **New standalone repo**, pulling from iRacing X3D Tuner + X3D Undervolt Tester; cross-linked both directions.
2. ~~Audience floor~~ → **Total first-timer**; one line on what BIOS is + access keys, then proceed.
3. ~~GO teaching path~~ → **All-core conservative default; opt-in Tuning Path** with software suite + logging + dual-format (human/AI) outputs.
4. ~~Vendor depth~~ → **Top-5 vendor translation table** with board-select rendering + "other terminology" dropdowns per setting.

Still open:

5. Memory scope for v1: EXPO + integrity only, or include a manual-timings advanced chapter
6. Do we cover the Windows install/debloat itself or start post-install
7. Tool packaging: individual scripts (kit style) vs a single menu-driven launcher
8. X3D Turbo Mode / SMT-off game modes: document, recommend against, or ignore
9. Log format for the history file: JSON vs structured

text (JSON recommended – machine-parseable for the AI block)

10. Repo name (candidates: x3d-first-time-setup, x3d-foundation, x3d-baseline-guide)

11. 5th vendor for the translation table (Biostar assumed – or NZXT?)

12. Web delivery: static GitHub Pages like the iRacing kit, or does board-select/dropdown rendering justify a small JS layer

8. Version History

v0.1 – Initial baseline (2026-07-25)

First structuring of the brain dump into a working plan.
Established:

- Purpose statement + the four project principles (verify don't assume; repeatability over peak numbers; auto-detect then branch; tested or sourced)
- Chip framework: Class A (single-CCD) / B (dual-CCD single-cache) / C (dual-CCD dual-cache) architecture matrix with per-SKU profile schema; 9950X3D as primary target
- Phase 0 (BIOS foundation): AGESA baseline, EXPO + memory training integrity strategy (MCR disabled + Power Down disabled for guaranteed identical training every boot), trained-value fingerprinting, SoC

- ≤1.30V guardrail, PBO/CO conservative-start philosophy, scheduling-relevant settings, boot integrity checklist
- Phase 1 (Windows baseline): chipset driver stack / V-Cache Optimizer, Game Bar warning for Class B, Balanced power plan, NVIDIA driver hygiene, handoff point to iRacing kit
- Phase 2 (testing kit): validation matrix (TestMem5/Karhu/y-cruncher/CoreCycler/OCCT/idle soak/latency), six-tool build list (detect reuse, BIOS auditor, fingerprint compare, scheduling verifier, health report, backup integration), tiered Quick/Standard/Paranoid protocol
- Tagging system: [VALIDATED] / [RESEARCH] / [VERIFY] / [DECISION NEEDED]
- Asset intake list, 8 consolidated research items, 8 open decisions

v0.2 — Decisions 1–4 resolved (2026-07-25)

- **Distribution resolved:** new standalone repo; source projects expanded to TWO — iRacing X3D Tuner AND X3D Undervolt Tester (new asset added to intake list); cross-linking both directions
- **Audience resolved:** total first-timer; one-line BIOS explainer + access keys, verification tooling as the safety net

- **CO path resolved:** two-path structure — Path 1 all-core conservative default for everyone; Path 2 opt-in Tuning Path (software suite + recommendations + full logging), absorbing/extending the X3D Undervolt Tester
- **Vendor depth resolved:** top-5 vendor BIOS Translation Table (ASUS, MSI, Gigabyte, ASRock, Biostar-pending) with one-time board select (or WMI auto-detect), vendor-specific rendering of instructions, and per-setting "other terminology" dropdowns
- **NEW Section 4.4 — Dual-Format Output Spec (core feature):** every tool outputs Layer 1 human-readable report + Layer 2 self-contained "Format for AI Troubleshooting" block (embedded prompt header defining role/objective/what-to-solve, structured state dump, explicit ask) with append-only history logging so iterative AI tuning sessions carry full context
- Decisions list restructured into Resolved (1–4) / Still open (5–12); four new opens added: log format (#9), repo name (#10), 5th vendor (#11), web delivery static-vs-JS (#12)

Next (v0.3 planned)

- Asset intake pass: detect script, Undervolt Tester, kit files, troubleshooting notes → convert

[VERIFY] / [VALIDATED-DIRECTION] tags to
confirmed claims

- Resolve decisions 5–12
- Begin research items 1–8